

Preference-shaped media walls for taste memory

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Yum is a personal taste-memory wall for cars, food, alcohol, and K-pop girls, rather than a food-only media site. The system combines a static media archive, source-aware image metadata, category balancing, remote candidate discovery, optional model-assisted ranking, and cross-device preference memory into a wall that opens directly into images instead of a dashboard. The central design problem is to preserve a specific four-category taste contract while keeping the public page fast, visually continuous, and useful when optional services are unavailable. Here the system is described as a static-first, preference-aware media surface in which local fallback material, remote candidates, hidden-item memory, and source de-duplication jointly maintain variety and personal fit.

1 Introduction

Most media walls are either static collections or generic recommendation feeds. Static collections preserve intent, but they become repetitive. Generic feeds keep moving, but they usually lose the specific taste contract that made the page worth opening in the first place. Yum is designed for the middle ground: a personally shaped wall that starts from known-good material, keeps extending through candidate discovery, and remembers negative feedback across sessions.

The subject scope is intentionally explicit. Yum is for cars, food, alcohol, and K-pop girls. The value is in allowing those categories to coexist without one category taking over, without source repeats becoming obvious, and without the page becoming a control panel before it becomes a visual experience.

The public surface is a static GitHub Pages site at `yum.aolabs.io`. A Node service provides optional dynamic behavior: preference persistence, model-assisted curation, and K-pop candidate discovery. The page still functions when those services are not available, because the first requirement is that the wall opens quickly and remains visually useful.

2 Design requirements

2.1 Media-first entry

The first screen should be the wall itself. The interface avoids a heavy onboarding flow, filter dashboard, or explanatory landing page. Captions, source links, and controls are available, but they should not compete with the images.

2.2 Four-category taste contract

The feed must hold four subject areas at once: cars, food, alcohol, and K-pop girls. This requires a stricter contract than appending images to a masonry grid. The system needs category balance, source de-duplication, person and car-group spacing, and fallback material that still matches the intended taste when online sources are weak.

2.3 Preference memory

Yum uses hide actions as meaningful taste signals. Hidden source keys remove exact items. Hidden samples become stronger negative examples. Nearby visible items can become weaker positive examples because they were present at the time of the rejection and were not removed. This keeps the stored memory compact while still giving the ranking layer useful direction.

3 System architecture

Yum has three main layers: a public static wall, a small backend service, and a preference model that is mirrored between browser state and remote storage. The public wall is responsible for responsive masonry placement, tile shapes, source links, captions, visual grouping, lazy loading, and interaction state. Each tile carries more information than its visible image: a category, shape, source key, caption, source URL, focus hint, and sometimes person or group metadata.

The backend preserves the parts of the system that cannot be solved with a static page alone. It accepts and returns preference state, filters and ranks curation candidates, caches responses, and exposes K-pop candidate retrieval as a separate endpoint. This keeps the wall operational while still allowing the taste model to improve beyond a fixed bundle.

4 Feed contract

The feed contract is the heart of Yum. The wall should feel like one visual surface, but the categories have different failure modes. Cars need modern, tasteful reference images and should avoid stale

show-floor, traffic, SUV, old-car, or disliked model and colour patterns. Food needs cravings, memory, restaurants, meals, dishes, screenshots, and the kind of image that is worth saving for later. Alcohol needs clean table, bottle, drink, and nightlife-adjacent material without turning the whole feed into bar stock imagery. K-pop girl imagery needs to stay softer and more personal, avoiding heavy stage energy, noisy editorial frames, and low-quality portraits.

This contract is enforced by multiple mechanisms rather than one filter. Static pools guarantee a baseline. Blocked-term checks remove obvious misses. Source keys prevent exact repeats. Visual groups and person groups prevent one look from dominating the scroll. Recent placement history keeps the wall from forming blocks of the same category.

5 Curation and preference stack

Yum separates candidate generation from candidate selection. Candidate generation answers the question, “what could be shown next?” Candidate selection answers the harder question, “what should be shown next, given the current wall and what has already been rejected?”

The browser stores a local mirror of preference state so the interface can react immediately. The backend stores the same class of compact state in `preferences.json`: hidden keys, hidden samples, kept samples, a version number, and an update timestamp. This is enough to make the wall feel persistent across devices without storing a full behavioural history.

The persistence model favors small, explicit signals. A large opaque log would be harder to reason about and harder to repair. A compact preference object can be inspected, capped, synced, and applied directly to candidate selection.

6 Conclusion

Yum is a preference-shaped media wall for cars, food, alcohol, and K-pop girls. Its core contribution is not the presence of those categories alone, but the operational system around them: static-first rendering, source-aware metadata, category balance, candidate ranking, compact preference memory, and cross-device persistence. Those pieces allow a simple visual wall to behave like a durable taste

surface instead of a one-time image collection. 81

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No external scholarly references are cited in this manuscript. 83

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Table 1. Current Yum architecture. The system is intentionally static-first, with dynamic curation layered on top instead of required for initial rendering.

Layer	Implementation role
Public surface	GitHub Pages serves the wall, static assets, preview metadata, icon assets, and the paper link at <code>yum.aolabs.io</code> .
Backend service	A Node service exposes <code>/api/curate</code> , <code>/api/preferences</code> , and <code>/api/kpop-candidates</code> .
AI ranking	The curation endpoint can score image candidates using the configured OpenAI model and the current taste memory.
Persistence	Preference state is stored locally for immediate response and remotely for cross-device continuity.
Fallback behavior	Static seed pools and generated candidate templates keep the wall useful when network sources or model calls fail.

Table 2. Curation mechanisms. The table summarizes the mechanisms used to keep Yum from collapsing into a generic image feed.

Mechanism	Purpose
Static seed pools	Start the wall with known-good cars, food, alcohol, and K-pop girl media.
Blocked terms	Reject low-fit candidates before they reach the wall.
Hidden keys	Remove exact source items that the user has rejected.
Hidden samples	Store stronger negative examples for model-assisted ranking.
Kept samples	Preserve lightweight positive context from nearby non-hidden items.
Model ranking	Rank candidates against current taste rules and preference memory when the API is configured.
Source rotation	Keep the wall moving when one candidate source runs thin or stale.

Figure legends

No figures are included in this manuscript.

Methods

Static surface and media metadata

The public surface is generated from a static bundle containing the HTML shell, CSS, JavaScript wall renderer, icon assets, preview images, and paper files. Media candidates are represented as structured entries with image source, source URL, caption, category, tile shape, focus hint, and optional person or visual-group metadata.

Preference storage

Preference state is stored as a compact object containing hidden keys, hidden samples, kept samples, a version number, and an update timestamp. The browser keeps a local mirror for immediate interaction. The backend stores the same preference shape so that state can be restored on another device when the remote service is reachable.

Candidate ranking

Candidate ranking is static-first. Known-good seed media is available before any remote service responds. Optional curation requests can pass candidate metadata and preference samples to the configured model, after which blocked terms, source keys, category balance, and recent placement history are still applied before display.

Verification

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The manuscript PDF was generated directly from the LaTeX source using `pdflatex`. The deployed site was verified by checking that the live paper URL returned a PDF and that text extracted from the deployed PDF contained the expected four-category scope and Nature-style manuscript end matter.

Data availability

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No new research dataset was generated. The media metadata required to inspect the public wall is included in the deployed public JavaScript bundle and in the Yum application source. Third-party media remains subject to the licensing and availability terms of the original sources linked from the wall.

Code availability

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The application code and this LaTeX source are available in the Yum application repository at github.com/nalalalan/yum-app. The deployed LaTeX source is also served at yum.aolabs.io/yum-preference-shaped-media-wall.tex.

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Author contributions

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A.N.P. conceived the system, implemented the wall and backend, designed the curation and preference-memory behavior, prepared the manuscript, and maintains the public deployment.

Competing interests

A.N.P. created and operates AO Labs and Yum.

Additional information

Supplementary information is not included. Correspondence and requests for materials should be directed to Alan N. Pham through aolabs.io.